$(u \quad v + 1$ =

$(\alpha_1^2, \alpha_2^2, \sqrt{2}\alpha_1\alpha_2, \sqrt{2}\alpha_1, \sqrt{2}\alpha_2, 1)$

$\quad 1 \quad$

$= \varphi(u) \quad \varphi(v)$ Where: $\varphi(x) = (x_2)$ $\Rightarrow$ $\quad 1 \quad$

$a = (2)$ $b = (4)$

$(1 \;, \; 2 \;, \; \sqrt{2}\, 1\cdot 2, \; \sqrt{2}, \; 2\sqrt{2}, \; 1)$

$\quad 1 \quad$

$\varphi(a) \quad \varphi(b) =$

$= (1, \; 4, \; 2\sqrt{2}, \; \sqrt{2}, \; 2\sqrt{2}, \; 1)$ $\quad 1 \quad$

$= 9 + 64 + 48 + 6 + 16 + 1 = 144$

$(a \quad b + 1) = (1 + 3 + 2\cdot 4 + 1) = 12 = 144$